Portfolio Construction and Trading Strategy

The objective of this project was to develop and test a monthly long-short trading strategy using price momentum factors applied to the holdings of the Invesco QQQ ETF. QQQ was selected because it contains a sufficiently large collection of actively traded stocks and provides a relevant ETF benchmark for evaluating the strategy.

Historical price and volume data were collected for QQQ and its constituent stocks. To avoid using information that would not have been available at the time of portfolio formation, the factor calculations were based on prices and volume shifted backward by 20 trading days. The portfolio was then restructured at the end of each month, and its performance was evaluated using the following month's realized returns.

Five momentum-related factors were calculated for each stock:

1. Slope of the 52-week price trend line.
2. Percentage above the 260-day low.
3. 4-week versus 52-week price oscillator.
4. 39-week historical return.
5. 51-week volume-price trend measure.

Because these factors are expressed in different units, each factor was standardized across the available stocks each month using a cross-sectional z-score. The five standardized factor values were then averaged to create a combined monthly momentum score for each stock.

At each portfolio formation date, the 15 stocks with the highest combined z-scores were placed in the long basket, while the 15 stocks with the lowest combined z-scores were placed in the short basket. The basic long-short portfolio return was calculated as the average return of the long basket minus the average return of the short basket. Therefore, a positive portfolio return indicates that the selected long stocks performed better than the selected short stocks during the following month.

Factor Selection Discussion

The momentum factors were intended to capture different parts of the same general trading idea: stocks with strong recent performance and favorable trends may continue to outperform weaker stocks over shorter investment horizons. However, the factor-correlation analysis showed that several of the price-based variables were closely related.

In particular, the 52-week trend measure, the 4/52-week oscillator, and the 39-week return measure were highly correlated with one another. This result is reasonable because all three variables are based primarily on recent price movement. The volume-price trend factor had noticeably lower correlations with the other factors, suggesting that it contributed somewhat different information by incorporating trading volume.

This finding is important because combining several highly related variables may unintentionally place too much emphasis on the same momentum signal. In this project, the five factors were retained in the final model in order to evaluate the complete Berry Cox momentum framework used in the code. However, a future improvement would be to reduce the weight placed on strongly correlated price factors or test alternative combinations of factors.

Equal-Weighted Long-Short Backtest Results

The first version of the strategy used equal weights within both the long and short baskets. Every selected long stock received the same weight within the long side, and every selected short stock received the same weight within the short side.

Over the valid backtest period, the equal-weighted long-short portfolio generated an average monthly return of **1.05%**, compared with an average monthly return of **1.80%** for QQQ. The portfolio also experienced greater monthly volatility, with a standard deviation of **7.91%**, compared with **5.95%** for the ETF.

Measure	***Equal-Weighted Long-Short Portfolio***	***QQQ ETF***
Number of Valid Months	70	70
Average Monthly Return	1.05%	1.80%
Monthly Standard Deviation	7.91%	5.95%
Minimum Monthly Return	-23.52%	-13.60%
Median Monthly Return	1.12%	1.91%
Maximum Monthly Return	22.41%	14.97%

The equal-weighted strategy produced several months of strong positive performance, including returns above those of QQQ during some periods. However, it also experienced larger negative months. Its worst monthly return was approximately **-23.52%**, substantially below QQQ’s worst monthly return of approximately **-13.60%**.

Using monthly returns for a simple annualized comparison, the portfolio produced an annualized average return of approximately **12.61%**, while QQQ produced approximately **21.60%**. Annualized volatility was approximately **27.39%** for the long-short portfolio compared with **20.61%** for QQQ. These results show that the equal-weighted strategy generated positive returns, but it did not outperform the ETF benchmark on either return or risk-adjusted performance during the sample period.

Interpretation of Equal-Weighted Charts

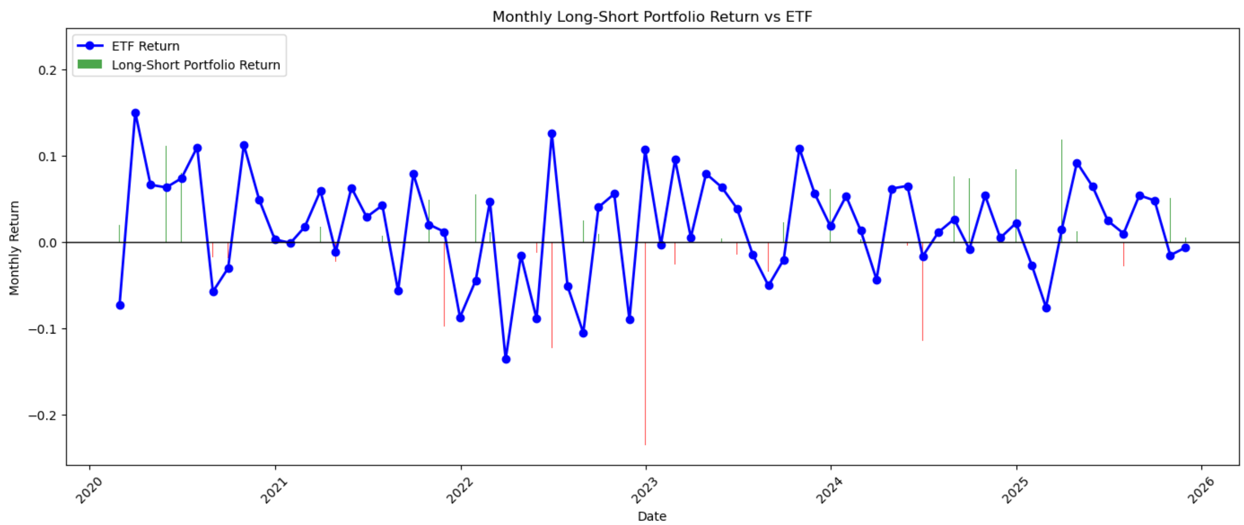

Figure 1: Monthly Long-Short Portfolio Return versus QQQ

The monthly return chart compares the long-short portfolio return with the monthly return of QQQ. Positive portfolio months are displayed in green, while negative portfolio months are displayed in red. The chart shows that the strategy produced both sizable gains and sizable losses. The portfolio was especially volatile during parts of 2022 and early 2023, when several negative portfolio months occurred even though QQQ later recovered.

The ETF generally followed a smoother pattern than the long-short portfolio. This suggests that the momentum-based stock selection process introduced additional stock-selection risk beyond simply holding the benchmark ETF.

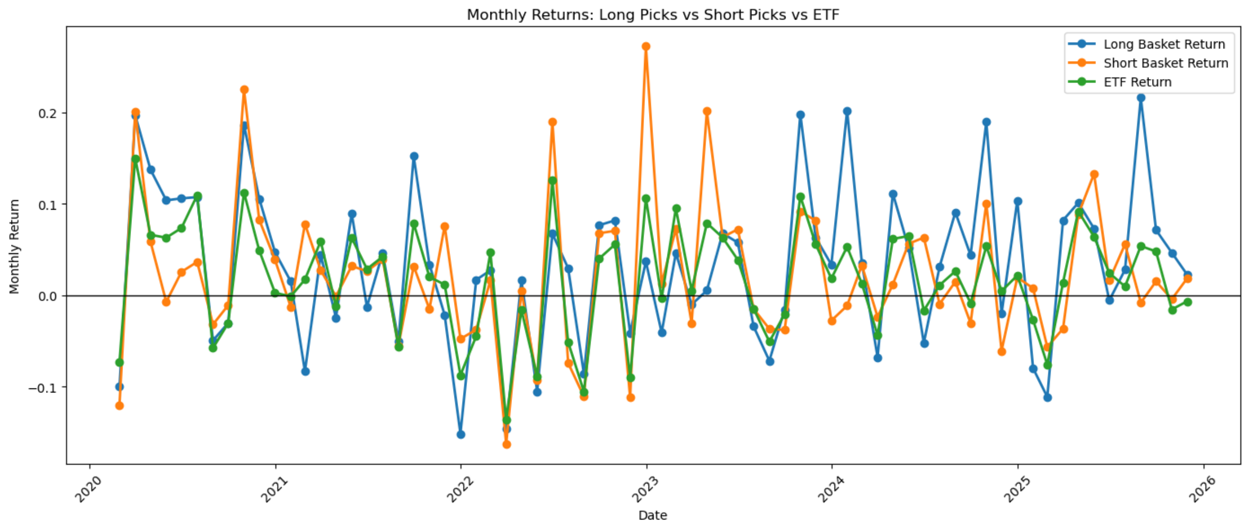

Figure 2: Monthly Returns of the Long Basket, Short Basket, and QQQ

The second chart separates the performance of the long and short baskets. For the long-short strategy to perform well, the long basket must outperform the short basket. The chart shows that this happened in several months, but not consistently throughout the entire period.

In some months, the short basket produced large positive returns. Since the strategy is short these stocks, such months negatively affected total portfolio performance. This is an important reason why the strategy sometimes underperformed QQQ even when the selected long stocks produced positive returns.

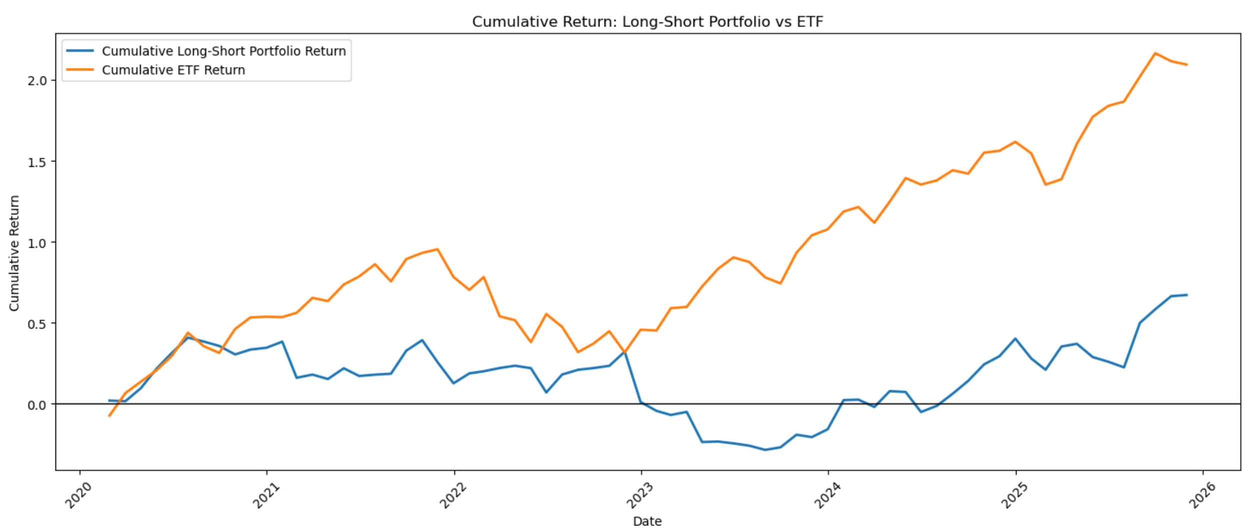

Figure 3: Cumulative Return of the Equal-Weighted Long-Short Portfolio versus QQQ

The cumulative return chart provides the clearest overall comparison. The equal-weighted long-short strategy initially produced positive results and remained competitive with QQQ during part of the earlier sample period. However, the portfolio experienced a major decline around 2022 and early 2023. Although it later recovered, its cumulative performance remained far below that of QQQ by the end of the backtest.

By the end of the period, QQQ had generated a cumulative return of more than 200%, while the equal-weighted long-short strategy ended with a cumulative return of approximately 65% to 70%. Therefore, the momentum strategy produced a positive cumulative return, but passive investment in the ETF would have produced substantially better results during this period.

Optimized Weighting Method

For the extra-credit portion of the project, I replaced equal weighting within the selected baskets with optimized monthly weights. The stock selection process did not change: each month, the 15 highest-ranked stocks remained in the long basket and the 15 lowest-ranked stocks remained in the short basket.

The difference was that the selected stocks were no longer automatically weighted equally. For each monthly restructuring date, a Monte Carlo portfolio optimization procedure was applied using the prior 12 months of monthly stock returns.

For the long basket, the optimization selected the simulated portfolio with the highest estimated Sharpe ratio. This approach attempted to concentrate the long side in stocks that had offered a more favorable historical risk-return tradeoff.

For the short basket, the optimization selected the simulated portfolio with the lowest historical volatility. This was intended to avoid creating an excessively unstable short position while still shorting stocks selected by the momentum ranking process.

The optimization was performed again each month when the portfolio was restructured. This prevented future information from being used in the weighting process because only information available before the following holding month was used.

Optimized Portfolio Results

After removing the final month that did not have a valid following-month ETF return, the optimized portfolio remained broadly similar to the equal-weighted portfolio. Based on the aligned observations, the optimized strategy generated an average monthly return of approximately **1.03%**, with monthly volatility of approximately **7.82%**. Its worst monthly result was approximately **-18.41%**.

Measure	***Equal-Weighted Portfolio***	***Optimized Portfolio***	***QQQ ETF***
Number of Valid Months	70	70	70
Average Monthly Return	1.05%	Approximately 1.03%	1.80%
Monthly Standard Deviation	7.91%	Approximately 7.82%	5.95%
Minimum Monthly Return	-23.52%	-18.41%	-13.60%
Median Monthly Return	1.12%	Approximately 2.20%	1.91%
Maximum Monthly Return	22.41%	20.78%	14.97%

The optimized portfolio slightly reduced monthly volatility relative to the equal-weighted strategy. It also reduced the size of the worst monthly loss from approximately **-23.52%** to **-18.41%**. This suggests that monthly weighting optimization provided some downside protection.

However, the optimized portfolio did not improve average monthly return. Its average return remained slightly below that of the equal-weighted version and substantially below that of QQQ. The reduction in volatility was also relatively small. Therefore, the optimization procedure improved some risk characteristics, but it did not materially improve the strategy's overall performance.

Using a simple annualized comparison, the optimized portfolio generated an annualized average return of approximately **12.34%** and annualized volatility of approximately **27.09%** after aligning the valid months. This remained below QQQ's approximately **21.60%** annualized average return and above QQQ's approximately **20.61%** annualized volatility.

Interpretation of Optimized Charts

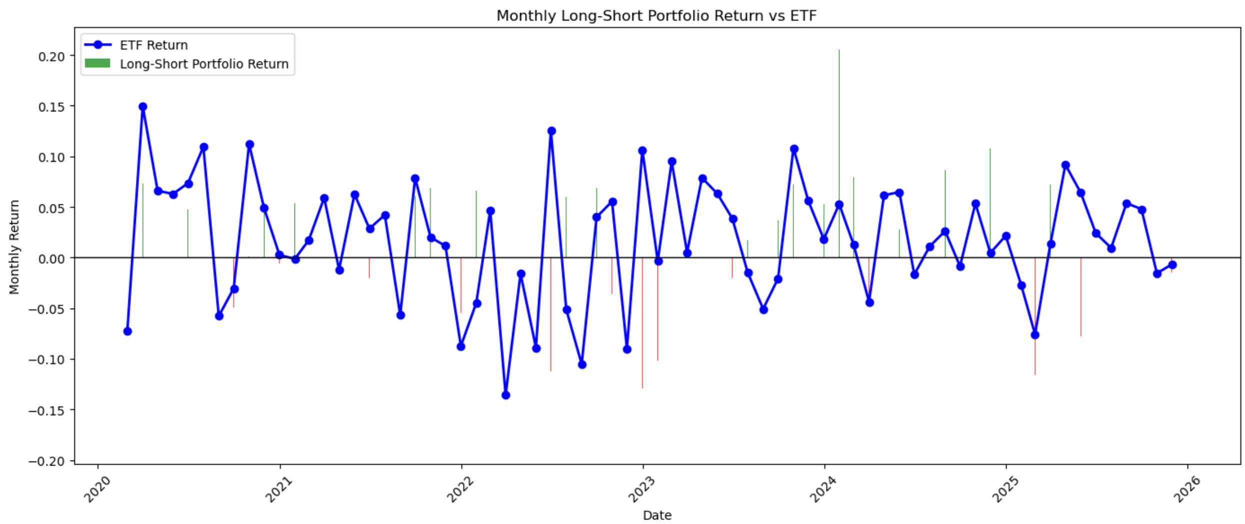

Figure 4: Monthly Optimized Long-Short Portfolio Return versus QQQ

The optimized monthly return chart shows that optimized weighting did not eliminate negative portfolio months. The strategy continued to experience losses during periods when the short basket performed strongly or when the long basket underperformed.

Compared with the equal-weighted chart, the optimized portfolio avoided an extremely large loss of the same magnitude as the equal-weighted portfolio's worst month. Nevertheless, the monthly returns remained more volatile than those of QQQ.

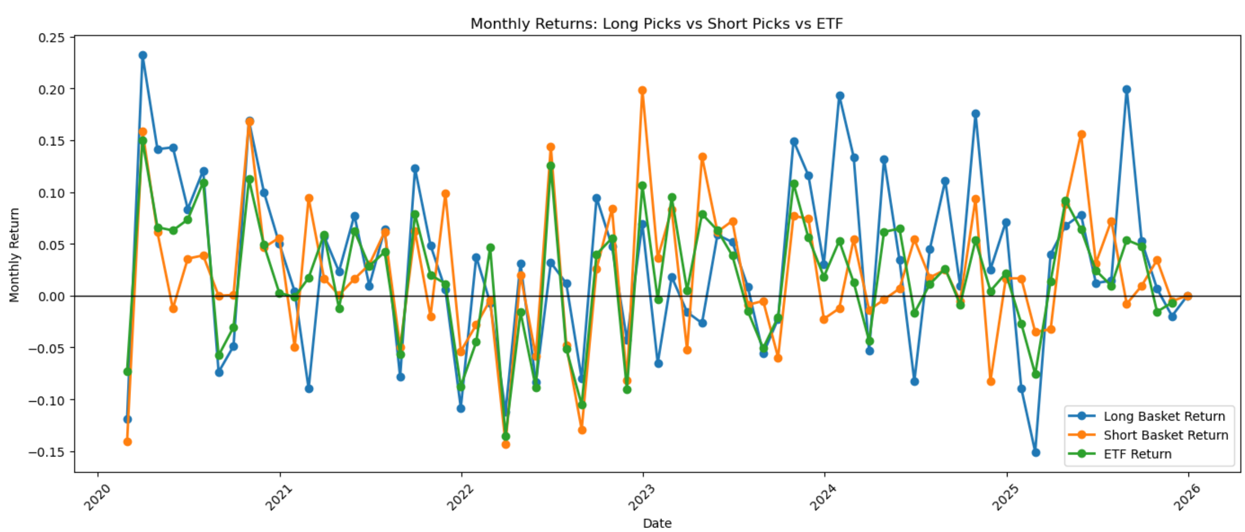

Figure 5: Monthly Returns of the Optimized Long Basket, Short Basket, and QQQ

The optimized long and short basket chart shows that the weighting process changed the size of several monthly returns, but the overall pattern remained similar. The long basket occasionally generated very strong positive results, but the short basket also experienced positive surges that worked against the strategy.

This result indicates that weighting alone could not solve the main weakness of the strategy. The more important challenge was whether the z-score ranking method correctly distinguished future winners from future losers.

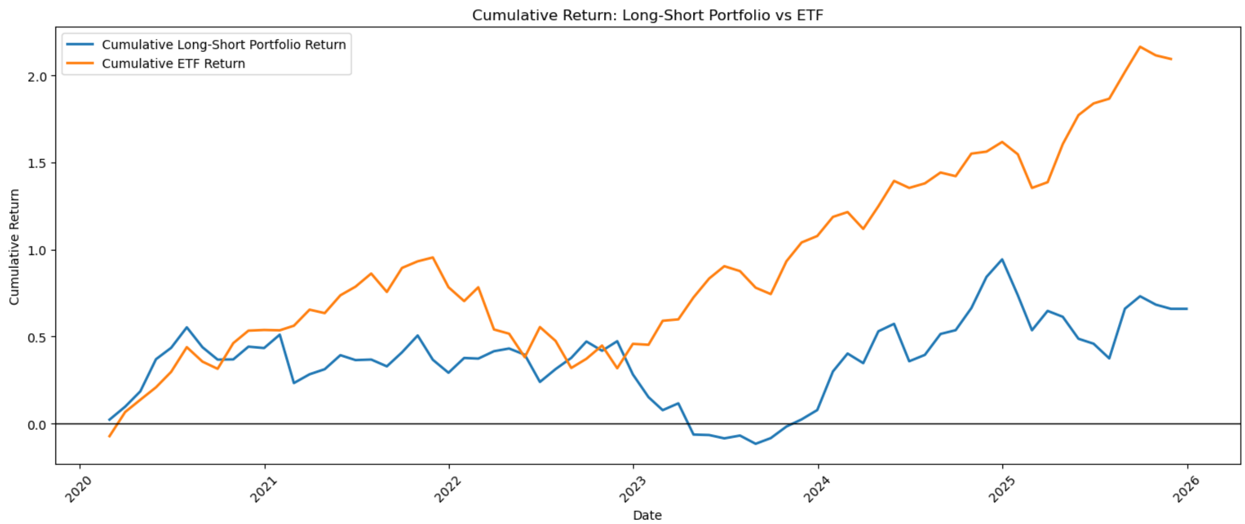

Figure 6: Cumulative Return of the Optimized Long-Short Portfolio versus QQQ

The optimized cumulative return chart shows that optimized weighting produced some improvement during parts of the backtest, particularly during the later recovery period. At its strongest point, the optimized strategy reached a higher cumulative value than the equal-weighted strategy.

However, QQQ still finished far ahead of the optimized strategy. The optimized long-short portfolio ended with cumulative gains of approximately 65% to 70%, while QQQ finished above 200%. Thus, optimization improved the behavior of the portfolio slightly, but it did not change the overall conclusion that the benchmark ETF performed better over the tested period.

Comparison of Equal Weighting and Optimized Weighting

The comparison between the two portfolio methods provides an important practical result. Optimization did not automatically produce superior performance. Although the optimized method reduced the worst monthly loss and slightly lowered volatility, it also produced a slightly lower average monthly return.

Comparison	***Result***

Higher Average Monthly Return	Equal-Weighted Portfolio
Lower Monthly Volatility	Optimized Portfolio
Smaller Worst Monthly Loss	Optimized Portfolio
Higher Benchmark Return	QQQ ETF
Best Overall Performance During Sample	QQQ ETF

This result is reasonable because the optimizer can only change the weights of stocks already selected by the momentum ranking model. If the factor model selects weak long positions or short positions that later rise in value, optimized weighting may reduce the damage but cannot completely eliminate the problem.

In addition, the optimization relied on only a 12-month historical return window. A short estimation window can cause portfolio weights to react strongly to recent returns that may not persist in the following month. As a result, the optimized weights may have reduced diversification without consistently improving future performance.

Limitations

Several limitations should be considered when interpreting the results.

First, many of the selected momentum factors were highly correlated. The trend, oscillator, and historical return measures were all based on recent price movement. Combining them equally may have caused the model to repeatedly emphasize similar information rather than independent signals.

Second, the backtest was conducted using the current constituent list of QQQ. This creates the possibility of survivorship bias because stocks currently included in the ETF may not perfectly represent the stocks that were included at every historical point in the backtest period.

Third, the analysis does not include transaction costs, short-selling costs, bid-ask spreads, borrowing constraints, or taxes. Because the portfolio was restructured monthly and held both long and short positions, these costs could meaningfully reduce realized performance.

Fourth, the optimization procedure was based on recent historical returns and covariances. Historical patterns may not continue in the following month, particularly during changing market conditions. The optimized weights should therefore be interpreted as a test of an additional allocation method rather than proof that optimization creates superior returns.

Finally, the strong performance of QQQ over the tested period created a difficult benchmark for a market-neutral long-short strategy. A long-short portfolio is designed to profit from differences between selected stocks, while an ETF benefits directly from a rising overall market. Therefore, comparison with QQQ is useful, but the strategies do not have identical market exposure.

Conclusion

This project tested a monthly long-short portfolio strategy based on Berry Cox price momentum factors applied to the stocks in QQQ. The strategy ranked stocks each month using standardized momentum scores, purchased the 15 strongest-ranked stocks, and shorted the 15 weakest-ranked stocks. The resulting portfolio was evaluated using following-month returns over approximately five years of valid monthly observations.

The equal-weighted long-short portfolio generated positive cumulative performance, showing that the momentum rankings produced profitable opportunities during some parts of the sample. However, the strategy did not outperform QQQ. The equal-weighted portfolio generated lower average returns and greater volatility than the benchmark ETF.

The optimized weighting extension slightly improved downside risk by reducing the largest monthly portfolio loss and lowering volatility. However, it did not increase average returns or close the cumulative performance gap relative to QQQ. This suggests that the central issue was not simply how selected stocks were weighted. Rather, the more important issue was the predictive strength and redundancy of the momentum factors used to select the long and short baskets.

Overall, the results show that price momentum factors can be used to construct a systematic and testable trading algorithm, but positive signals in historical data do not guarantee benchmark outperformance. A stronger future version of the model would test less-correlated factor combinations, include transaction and shorting costs, use historical ETF membership where available, and evaluate whether alternative weighting approaches improve performance out of sample.